

JUSTIFICATION
FOR AN EXCEPTION TO FAIR OPPORTUNITY

1. Contracting Activity: Department of Veterans Affairs (VA)
Office of Procurement, Acquisition, and Logistics
Technology Acquisition Center
23 Christopher Way
Eatontown, NJ 07724
2. Description of Action: The proposed action is for a firm-fixed-price delivery order issued under the National Aeronautics and Space Administration (NASA) Solutions for Enterprise-Wide Procurement (SEWP) V Government-wide Acquisition Contract (GWAC) for brand name Innovative Research TileFlow (TileFlow) Computational Fluid Dynamics software licenses and maintenance.
3. Description of the Supplies or Services: The VA Office of Information and Technology (OIT), Data Center & Infrastructure Engineering (DCIE) Team has a requirement for software, upgrades, and technical product support for the TileFlow Computational Fluid Dynamics (CFD) software platform. The VA requires the purchase of four brand name TileFlow software licenses, unlimited product support, and software updates. The VA also requires a robust CFD platform capable of reading previously created TileFlow CFD models, supporting centralized on-demand floating licensing for up to four users and obtaining/maintaining VA Technical Reference Model (TRM) approval. The required product must enable DCIE to continue providing data center and telecommunications space design and engineering assessments in support of new Electronic Health Record Modernization facilities, as well as modifications to existing spaces, to ensure optimal and cost-effective thermal performance.

The proposed action is for a Service-Disabled Veteran-Owned Small Business (SDVOSB) set-aside. The total estimated value of this action for the software licenses and maintenance support to be purchased with a 12-month base period and four 12-month option periods is \$ [REDACTED]

4. Statutory Authority: The following brand name requirement is being justified and approved as an exception to fair opportunity under the statutory authority of Section 41 U.S.C. 4106(c)(2) as implemented by Federal Acquisition Regulation (FAR) Overhaul VA Deviation 16.507-6(b)(2), entitled "Only one awardee is capable of providing the supplies or services required at the level of quality required because the supplies or services ordered are unique or highly specialized."

5. Rationale Supporting Use of Authority Cited Above: OIT initially purchased TileFlow software licenses in 2017 via NASA SEWP to support simulation of thermal performance within VA Data Centers. The TileFlow CFD software platform is used by VA engineering staff within OIT and Veterans Health Administration, Healthcare Environment and Facilities Programs, to design and analyze data center environments, plan IT equipment deployments to avoid adverse impacts to thermal conditions, and assess remaining capacity at field stations. VA engineering staff utilizing the tool

operate at the national level or directly support OIT Enterprise Data Centers regionally. The software enables enterprise-level analysis and planning efforts supporting approximately 170 data centers nationwide.

Market research confirms that TileFlow is the only software capable of meeting VA's requirements while leveraging and extending VA's existing catalog of CFD analyses. TileFlow meets several critical requirements, including lightweight standalone architecture that enables easy deployment and management across diverse Information Technology environments. Its intuitive Graphical User Interface (GUI) streamlines the creation and analysis of thermal models, improving user efficiency. TileFlow also provides unlimited technical support, ensuring that VA users have access to assistance whenever needed. The software's compatibility with legacy TileFlow models preserves the VA's prior investments, while its shared licensing model supports flexible, multi-user access to optimize resource allocation. The software includes a library of predefined objects that accelerates model development and analysis.

Further, TileFlow is listed as an authorized software within the VA TRM, confirming compliance with the technical standards required for integration into VA's Enterprise Architecture. TileFlow remains the only CFD software platform approved for VA use for modeling airflow and heat loading within VA data center environments. Any alternative CFD software would be required to undergo the VA TRM approval process prior to implementation. The TRM approval process is a rigorous, multi-step process involving technical, security, and 508 compliance reviews and may require several months to over a year to complete. As a result, adoption of an alternative CFD software platform introduces significant delays and administrative burden before the alternative software could be authorized for operational use within the VA.

CFD models supporting hundreds of VA facilities and standards models use the proprietary TileFlow (.TLF) file extension. This file type is not compatible with competitor CFD software platforms, and no known systems exist that can convert this proprietary file type into another without wholesale re-creation and re-engineering of the models. Over the past nine years, VA has developed and validated data center thermal performance models within the existing TileFlow environment. Transitioning to an alternative CFD platform would require the redevelopment, testing, and validation of these existing models. The re-creation effort is estimated to require approximately [REDACTED] labor hours by personnel fully trained in both replacement software and VA engineering standards requirements at an estimated cost of [REDACTED]. This estimated cost is based on a not-to-exceed, fully burdened median engineering labor rate of [REDACTED], which is derived from the General Services Administration Pricing Intelligence Suite, Contract Awarded Labor Category + Quick Rate tool and reflects the work being performed by an engineer possessing a master's degree and approximately 15 years of experience, performing work at the contractor's facility under a small business socio-economic category within the information technology sector for fiscal

year 2026. Such an effort would result in substantial duplication of cost, which would not be recouped through competition.

Additionally, VA Subject Matter Experts (SMEs) have established standardized configuration models and repeatable processes around TileFlow that produce consistent, high-confidence modeling results for both raised floor and non-raised floor environments, whether in As Planned or As Built conditions. Their deep expertise in configuring and operating TileFlow significantly reduces the effort required to generate accurate and reliable models. These models are actively used to plan, validate, and maintain airflow and thermal performance across VA data centers. Continued support for legacy TileFlow models is therefore essential to maintaining operational continuity and ensuring consistency in ongoing and future analyses.

Transitioning to an alternative CFD software platform would require the VA to re-establish validated modeling methodologies and rebuild expertise developed over nearly a decade of operational use. Such a transition would disrupt standard configuration models and established analytical processes that VA SMEs currently rely upon, potentially introducing inefficiencies and increased uncertainty in thermal performance simulations. The overall Government impact would extend beyond initial transition activities and include additional costs and resources associated with transitioning to a new CFD software platform. These efforts would include, but not limited to, the time and resources required to convert existing CFD analyses to the new CFD software platform, train personnel, recalibrate models, validate the accuracy of new software outputs, and manage the potential delays in ongoing projects resulting from the learning curve associated with using a new tool.

In summary, the financial and operational burdens of switching to another software at the current time significantly outweighs the perceived benefits, making TileFlow the most viable and cost-effective option for meeting VA's requirements.

6. Efforts to Obtain Competition: Market research was conducted, details of which are in the market research section of this document. Given the highly specialized nature of this software, this effort did not yield any additional sources that can meet the Government's requirements. However, the undersigned VA technical representative determined that limited competition is available among authorized resellers for brand name licenses, as well as software maintenance and support. In accordance with FAR Overhaul VA Deviation 5.301 and 16.507-6(f), the award notice for this action will be synopsized on the Contracting Opportunities Page and the justification will be made publicly available within 14 days of award.

7. Actions to Increase Competition: The Government will continue to conduct market research to ascertain whether there are changes in the marketplace that would enable future actions to be completed.

8. Market Research: Market research for this requirement has been conducted on an ongoing basis since 2017, with the most recent update occurring in December 2025. VA Data Center SMEs regularly reviewed industry trade publications and conducted internet research to ascertain if any other brand name software is available. Three vendors with similar software were reviewed - CoolSim, EcoStruxure IT Design CFD, and Cadence DataCenter Design Software. Based on reviews of these products, the Government's technical experts determined that none of these products can meet the Government's CFD functionality across all the following requirements: lightweight standalone architecture, intuitive GUI, unlimited technical support, full compatibility for legacy TileFlow models, shared licensing model, built-in library of predefined objects, and TRM approval. While CoolSim and EcoStruxure each provide an intuitive GUI and a library of predefined objects, both lack standalone architecture, unlimited support, legacy-model compatibility, shared licensing, and TRM approval. Cadence supports only a library of predefined objects; however, it does not meet the VA's requirement for the remaining requirements. Based on all these market research efforts, the Government's technical experts have determined that only Innovative Research brand name software can meet all the Government's requirements.

The Small Business Administration (SBA) certification database on the SBA Veteran Small Business Certification Program (VetCert) portal at <https://veterans.certify.sba.gov/> was searched using North American Industry Classification System (NAICS) code 541519, Other Computer Related Services, on December 3, 2025 and yielded 52 SDVOSB and VOSB vendors that can potentially provide the required supplies and services under that NAICS code. Additionally, the Contract Specialist conducted further research using the Provider Look-up tool on the NASA SEWP V GWAC website to ascertain the extent of limited competition among resellers of the brand name TileFlow software licenses. The Provider Look-Up found a total of 27 contract holders of which seven were documented as SDVOSB that can provide required brand name TileFlow software and maintenance.

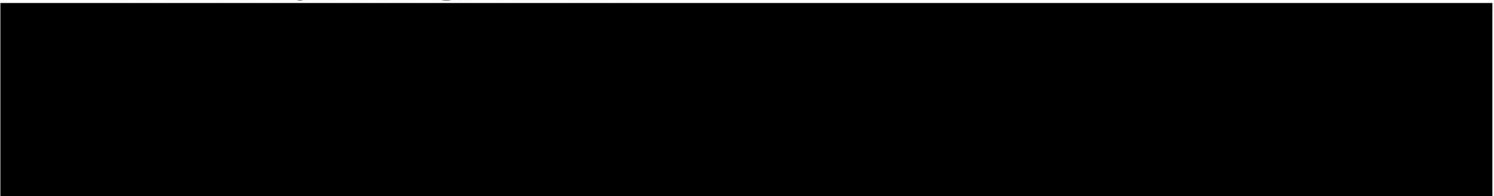
The TAC issued a Sources Sought on January 2, 2025, on NASA SEWP and the Contract Opportunities Page via SAM.gov. The Sources Sought included the request for brand name TileFlow software licenses and maintenance. A total of four responses were received by one Small Business and three SDVOSBs. All capability statements were reviewed, and no company disputed or objected to TileFlow licenses as the brand name software license required to meet the Government's needs outlined in the Product Description. Market research indicates that there are two or more SDVOSBs on NASA SEWP capable of meeting the Government's requirements.

Therefore, to promote award opportunity for SDVOSB concerns, the VA intends to issue a Request for Quote (RFQ) via NASA SEWP Quote Request Tool. To meet VA's socioeconomic goals, this RFQ will be issued as a SDVOSB set-aside among NASA SEWP GWAC group under the Value-Added Reseller NAICS code 541519 Group B,

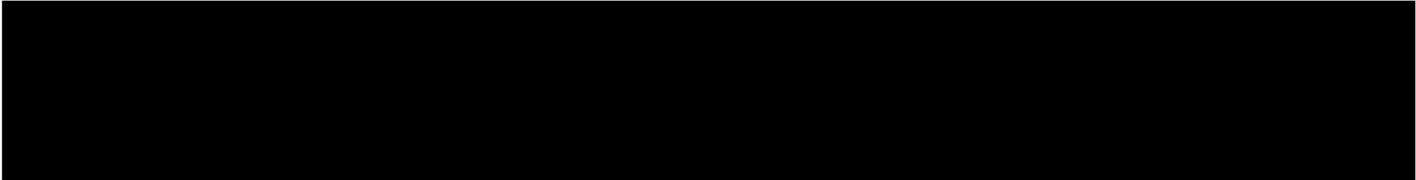
"SDVOSB Awarded as SDVSOSB Set-Asides." There are currently at minimum seven SDVOSB NASA SEWP contract holders that could compete for this effort.

9. Other Facts: None.

10. Technical and Requirements Certification: I certify that the supporting data under my cognizance, which are included in this justification, are accurate and complete to the best of my knowledge and belief.



11. Fair and Reasonable Cost Determination: I hereby determine that the anticipated price to the Government for this contract action will be fair and reasonable based on historical pricing and a comparison with the Independent Government Cost Estimate.



12. Procuring Contracting Officer Certification: I certify that this justification is accurate and complete to the best of my knowledge and belief.

